

Nandhini Dhanasekaran

206-530-4780 | ndhanu16@gmail.com | [linkedin/nandhinidhanasekaran](https://linkedin.com/in/nandhinidhanasekaran) | github.com/NandhiniDhanu

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer

Radiance Learning

Apr. 2025 – Present

Bellevue, WA

- Engineered a full-stack education platform with role-based access control for users enabling structured task delegation, secure user separation, and streamlined collaboration across user types.
- Developed 15+ RESTful APIs using Node.js (MongoDB) and Flask (PostgreSQL) for teacher/admin/parental messaging features, deployed via Azure App Services, ensuring high availability and protection of sensitive student/teacher data.
- Designed a responsive Vue.js admin dashboard with interactive task views, drag-and-drop sticky notes, dynamic breadcrumbs, and conditional data rendering, improving administrative efficiency and enabling real-time oversight of student and teacher activity.
- Created and maintained MongoDB and PostgreSQL schemas with clean relational mappings and ensured full-stack integration, reducing cross-service latency by 30% with schema alignment for real-time synchronization.
- Automated CI/CD pipelines with GitHub Actions and Azure deployment slots, incorporating linting, environment-based rollouts and unit tests, which reduced release time and improved delivery consistency.
- Facilitated sprint planning, design reviews, and API-UX alignment efforts, resulting in faster turnaround on frontend bugs and smoother cross-team integration.

Software Engineer

Health Chain

Jun. 2024 – Apr. 2025

Plano, TX

- Built an enterprise-grade Consent Management Portal using Next.js, TypeScript, and Tailwind, adhering to HIPAA standards, enhancing accessibility and regulatory compliance
- Implemented secure backend integrations with RBAC, token auth, and audit logging, ensuring HIPAA-aligned data access and reducing unauthorized access attempts by over 95%.
- Improved API performance using React Query, cutting average latency by 20% and improving cache hit rates via async architecture.
- Participated in architecture/code reviews on Azure-based deployments, improving integration quality and post-deployment defects.

Software Development Engineer (SDET)

Infosys Limited

Jul. 2020 – Aug. 2022

Chennai, IN

- Executed UAT test cases (Java, Selenium, Maven) for 30+ microservices in a financial platform, improving service uptime by 30%.
- Designed reusable test frameworks, reducing regression testing time by 50% and accelerating Agile sprint cycles.
- Scrutinized 500+ production logs weekly, pinpointing root causes of critical defects, leading to creation of 200+ targeted test cases.

PROJECTS

CodeOrbit – Cloud-Based Collaborative IDE | *React.js, Node.js, Docker, AWS EC2, Monaco, Socket.IO*

May. 2025

- Architected and deployed a real-time collaborative code editor via Socket.IO rooms, enabling multi-user sessions with auto state recovery and 150 ms latency.
- Engineered secure, language-agnostic execution environments (Python, JavaScript, Java) using Docker containers on AWS EC2, enabling scalable and isolated runtime handling.
- Automated deployments with GitHub Actions to EC2, establishing rollback-safe CI/CD workflows and 40% faster deployment.
- Applied fault-tolerant state syncing and reconnect logic, ensuring seamless UX and real-time recovery after disconnects.

AWS TLQ Data Pipeline | *Java, AWS Lambda, AWS EC2, AWS API Gateway, Amazon S3, SQLite*

Nov. 2022

- Architected a real-time serverless data processing pipeline using Java and AWS services including Lambda, S3, API Gateway, and EC2 for ingesting high-volume telemetry.
- Applied transformations and filtering using SQLite, reducing in-memory processing loads and overall runtime costs by 45%.
- Enforced robust API gateway security, including token-based access and throttling, while maintaining low-latency service access.

EDUCATION

University of Washington

Master of Science in Computer Science and Systems

Sep. 2022 – Mar. 2024

Seattle, WA

Anna University

Bachelors in Electronics and Communication Engineering

Nov. 2016 – Apr. 2020

Chennai, IN

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, Typescript, HTML5/CSS

Frameworks and Tools: React.js, Vue.js, node.js, Next.js, Flask, CI/CD, Springboot, Pandas, NumPy, Matplotlib, NLTK, Spark, Seaborn, MapReduce, Scikit-learn, Selenium, Postman

Cloud Services: Amazon Web Services(AWS Lambda, AWS EC2, AWS API Gateway, Amazon S3,AWS CloudWatch, SQLite), Microsoft Azure

Databases: MongoDB, PostgreSQL

Skills: Data Structures and Algorithms, REST API, SDLC, Manual/Automation testing, Unit Testing, API Testing, Agile, Scrum

RESEARCH PUBLICATION

IoT Based Emergency Alert System (View Publication)

Published: Jan 2021

A novel multifaceted safety and health monitoring system integrating IoT to enhance road safety, women safety, and provide continuous health monitoring for the elderly.